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Assertions and lexical invisibility in EFL learners' academic essays

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Abstract

The two studies presented here analyzed English academic essays written by Japanese learners of English and native speakers of English. A corpus-basis analysis indicated that the Japanese learners of English used more boosters than hedges, using straightforward and explicit expressions in English, whereas native speakers of English tended to express themselves more tentatively, using more hedges than boosters. Furthermore, a follow-up study sought to discover whether the Japanese learners of English were aware of the important pragmatic roles of hedges and boosters. The quantitative results suggested that although the certainty items were more visible to the Japanese learners of English, the learners had greater difficulty in identifying the pragmatic roles of the possibility and probability items and misapprehended their meanings. The relationship between possibility and probability items and metaphors of spatial relations for EFL learners is also considered.
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1. Hedges and boosters: an overview of the concepts

Lakoff (1972) defined hedges as ''words whose job it is to make things fuzzier or less fuzzy'' (1972, p. 195). According to his basic concept, hedges modify words or phrases within a proposition and make them fuzzier or less fuzzy. After Lakoff's popularization of the concept of hedges, his theory attracted the attention of many scholars and inspired a significant amount of research (e.g., Brown and Levinson, 1987; Fraser, 2010). Brown and Levinson (1987) later defined the term hedge as ''a particle, word or phrase that modifies the degree of membership of a predicate or noun phrase in a set; it says of that membership that it is *partial* or true only in certain respects, or that it is *more* true and complete than perhaps might be expected'' (1987, p. 145). Like Lakoff, they considered hedges to have the function of both attenuation and reinforcement, although Fraser (2010) viewed hedges as having only the former role, labeling modifiers with the latter role as boosters. To date, the literature and research regarding hedges has generally focused more on the attenuation aspect (Fetzer, 2010).

The concept of hedging has moved far from its origins since the early 1970s and includes not only the modification of words or phrases within a proposition but also the modification of commitment to the propositional content. Writers express their judgments, opinions and commitments toward the propositional content of the text and toward their readers through the use of hedges or boosters, modifying the truth-value of the whole proposition. While hedging devices reduce

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the strength or force of an expression, communicating tentativeness and possibility, boosting devices intensify or emphasize the force, expressing conviction and asserting confidence in a proposition (Holmes, 1995). The original concepts of hedge and booster have been further broadened to include the approaches in which hedges and boosters are used to realize an interactional/communicative strategy. Hyland (2012) explained that hedges can offer a possibility for textual manipulation in that they are used not only as a strategy to gain readers' acceptance of the writers' provisional statements but also as the writers' invitation to the readers to get involved in open discussion about the nature of the propositions. Boosters, on the other hand, are used as a strategy not only to persuade the readers of the writers' confident assertion but also to restrict the negotiating space available to the readers. In the present study, a ''hedge'' is defined as a mitigating word which attenuates either full semantic value of a particular expression or the full force of a speech act, whereas a ''booster'' as an intensifying word which reinforces either semantic value of a particular expression or the full force of a speech act. In other words, a ''hedge'' modifies meaning by making the message less forceful, while a ''booster'' modifies meaning by making the message more forceful.

Learners' expressions of hedges and boosters indicate different degrees of cognitive ability. To compare the use of hedges and boosters by second language (L2) learners and native speakers, the present study used epistemic categories ranging from maximum to minimum certainty.

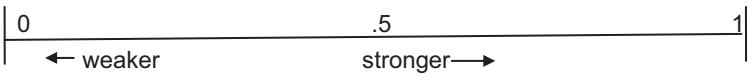
2. Categorizations of certainty, probability and possibility

The present study adopted the epistemic categories proposed by Halliday and Matthiessen (2014) and Hyland and Milton (1997) for adjectives and adverbs. In addition, to establish a more precise epistemic scale ranging from maximum to minimum certainty, the present study also utilized Horn's (2001) neg-raising theory for auxiliaries and verbs.

Horn (2001, p. 324) developed a meta-scale between zero and one, with zero being the weakest and one the strongest, incorporating both epistemic and deontic operations:

the weakest and one the strongest, incorporating both epistemic and deontic operations:

may, might	believe, suppose, think	know, realize
can, could	seem, appear, look like	must, have to
	should, ought to	



Lassiter (2011) considered verbs and auxiliaries at the weakest and strongest levels on the meta-scale to be minimum certainty and maximum certainty, respectively, whereas verbs and auxiliaries at the intermediate level are relative certainty because they are involved in neg-raising. According to Horn (2001, p. 308), neg-raising involves the ''availability of a lower-clause reading or understanding for a higher-clause negation,'' as shown in the following schema:

$$\text{Not}\,[\text{Pred}\,[\text{S}]] \Rightarrow \text{Pred}[\text{Not}\,[\text{S}]]$$

Auxiliaries and verbs occupying the mid-range position on the meta-scale are neg-raising triggers, with the classic example being the verb *think*.

a. James doesn't think Maria is here.
b. James thinks Maria isn't here.

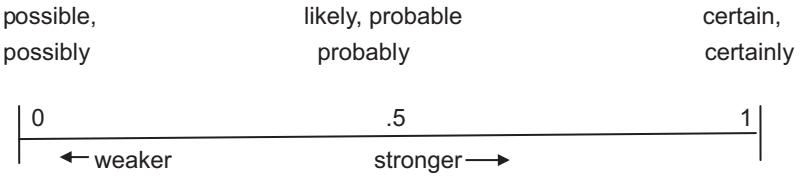
Intuitively, sentence (a) implies sentence (b). This relationship contrasts with the inferences associated with non neg-raising triggers at the weakest and strongest levels on the meta-scale.

a. *James doesn't know that Maria is here.
b. James knows that Maria isn't here.

In the above case, sentence (a) does not imply the corresponding sentence (b).

Neg-raising triggers are considered vague and lacking in sharp boundaries, reducing the strength or force of expressions and communicating tentativeness; they are also context-dependent (Lassiter, 2011). Lassiter (2010) semantically analyzed the epistemic adjectives such as *possible*, *probable*, *likely* and *certain* and argued that the

differences in the behavior of these adjectives are contingent upon their differing locations on a single scale of certainty, one that has both top and bottom elements. According to Lassiter (2010), *possible* is located at the weakest level, *probable* and *likely* at the intermediate level and *certain* at the strongest level on the meta-scale. In addition, because the meanings of epistemic adverbs are derived from the positive forms of their epistemic adjectives, epistemic adverbs can reasonably be classified in line with the classifications of their adjective forms on the meta-scale:



Accordingly, in the present study, auxiliaries, verbs, adjectives, and adverbs at the weakest level on the meta-scale are labeled as the possibility items, the ones at the intermediate level like the probability items, and the ones at the strongest level as the certainty items. The classifications of auxiliaries, verbs, adjectives, and adverbs on the meta-scale are related to the standpoint "from below" which is regarded as a lexical metaphor that takes the words as the starting point and then explains something about the meanings of these words.

3. Lexical metaphor and grammatical metaphor

According to conceptual metaphor theory (Lakoff and Johnson, 1999; Grady, 1999), humans use their understanding of the physical world as a framework for representing more abstract concepts. The lexical metaphor central to the present study is the proximal–distal metaphor in which certainty items are represented as being closer to the writer's confidence (e.g., She *will certainly* come here.), indicating a higher degree of sureness, whereas probability (e.g., She *would probably* come here.) and possibility items (e.g., She *might possibly* come here.) are differently represented as being further from the writer's confidence at different distances, indicating a lower degree of sureness. Tyler (2012) argued that the writer is much surer of the reality of things that he/she can perceive with his/her physical senses than the ones that are beyond the extent of his/her physical senses. This phenomenon means that when the writer feel more confident of propositions that he/she can experience or perceive with physical senses, certainty items will be used, and probability and possibility items will be used depending on how far propositions are from the writer's physical senses or how much confidence he/she has in propositions. Ohori (2002) and Ohori and Bo Sen (2012) explained that spatial relations are fundamental in human cognition and that spatial concepts are particularly effective sources of lexical metaphors. The lexical metaphor whose metaphorical nature is based on the use of a single word or lexeme and it is related to the writer's confidence in propositions in the present study, whereas a grammatical metaphor whose metaphorical nature is based on a grammatical construction type and it is associated with the writer's explicit indications of subjective–objective dimension about propositions in the present study.

The grammatical metaphor represents variation in the expression of meaning. Halliday and Matthiessen (2014) suggested that both the subjective explicit orientation (e.g., *I suppose* that. . .) and the objective explicit orientation (e.g., *It is likely* that. . .) are grounded in the grammatical metaphor because expressions of modality are extended from vocabulary to clauses by the addition of a projecting clause. Halliday and Matthiessen (2014) explained that using the subjective explicit orientation and the objective explicit orientation enables the writer to explicitly state the source of the conviction, but these orientations differ along the subjective–objective axis.

As for the subjective explicit orientation expressing certainty, the writer indicates that he/she alone knows the evidence and draws a conclusion assuming strictly personal responsibility for the epistemic judgment, whereas in the objective explicit orientation indicating certainty, the writer suggests that the evidence is clearly accessible to a number of people who share the conclusion, assuming shared responsibility for the judgment.

Table 1
Subjective/objective dimension of the grammatical metaphor.

		Subjective	Objective
Explicit	Certainty	I know that. . .	It is certain that. . .
	Probability	I suppose that. . .	It is probable that. . .
	Possibility	I assume that. . .	It is possible that. . .

Lexical and grammatical metaphors are of great importance in order to write academic papers effectively, and they are of concern not only to first language (L1) academic writers but also to writers who are learning English as a foreign language (EFL). The following section briefly touches on how L1 writers cope with hedges and boosters as lexical and grammatical metaphors briefly and examines how L2 learners address them in their writings.

4. Research on the use of hedges and boosters

A number of L1 studies in hedges and boosters have dealt with academic or scientific writing. Hedges are considered markers of uncertainty, and uncertainty is a fundamental characteristic of science writing. Grabe and Kaplan (1997) suggested that the motivation for expressing weak form negation with stronger pragmatic understanding involves the use of hedging or of politeness strategies. In addition, Hyland (1996) argued that in science writing hedges play a critical role in gaining approval for writers' claims from readers by presenting statements with appropriate accuracy, caution and humility. Examples of studies of hedges and boosters include medical English written discourse (Salager-Meyer, 1994), comparison of academic science writing and popular science writing (Crismore and Farnsworth, 1990), analysis of academic writing in the sciences, humanities and social sciences (Grabe and Kaplan, 1997; Hyland, 1998b), and analysis of modal verbs in science writing (Butler, 1990; Hanania and Akhtar, 1985). In sum, comparing hedges with boosters in L1 academic writing, hedges are more extensively used in academic articles and they are located densely in the discussion section. This reflects the importance of claims being presented provisionally rather than assertively to gain ratification from powerful peer readers.

Thus, writing academic papers provisionally is important to L1 academic writers, but it is also significant to EFL writers. Hyland (1998a) suggested that EFL learners find the correct and appropriate use of hedges to be extremely problematic, which may create obstacles to their participation in a research world dominated by English.

Clyne (1991) looked into a corpus of 52 texts—26 texts (17 texts in German and nine texts in English) by German scholars and 26 texts by English, American and Australian scholars. The disciplines examined were linguistics and sociology, and the texts by English scholars and German scholars were matched according to authors' sex, topic, length, disciplines, type of discourse, purpose and likely audience. The discourse structure indices were degree of linearity, symmetry, hierarchy of the text, continuity of text, presence and position of definition, functional sentence type, data integration and the relation between these indices. The results show that Germans used more hedging devices in academic texts than English counterparts irrespective of the language. Furthermore, the devices of modal verbs, impersonal constructions and passives were found more in texts by Germans regardless of language. Regarding the modal verbs, Germans authors tended to use *can* and *could*, whereas English authors accentuated the use of *may* (to a lesser extent, *might*). German authors even used double and triple hedging devices because they transferred the norms of their native culture when they wrote in English. As English is a West Germanic language and linguistically English and German are closely related, there is no denying that the short linguistic distance might have had spurring effects on the participants' use of hedging devices.

In contrast, Hyland and Milton (1997) examined the use of hedges and boosters by L2 learners whose native language was Chinese, linguistically distant from English, comparing English expressions of doubt and certainty in the writing of non-native speakers (NNSs) of English with such expressions in the writing of native speakers (NSs) of English. They also looked into how Chinese learners of English address the subjective/objective dimension of the grammatical metaphor in academic essays.

The 900 participants in their study included Cantonese-speaking high-school aged students and British students of a similar age and educational level. The data for the study consisted of two large corpora. The first corpus was a collection of essays written by Cantonese-speaking students for the high school matriculation General Certificate of English (GCE), *A* level, and totaled approximately 500,000 words, with 150 exam scripts in each of six ability bands. The grades on the exam ranged from *A* to *F* (Fail), with *A* being equivalent to a score of 600 on the TOEFL paper-based test and *E* comparable to a TOEFL score of approximately 450. The second corpus, also comprising 500,000 words, was transcribed from 770 GCE *A* level General Studies scripts written by British students. The authors reported that the NNSs used more boosting devices than the NSs, whereas the NSs employed more hedging devices. Among the NNSs, lower proficiency students used a significantly higher proportion of boosting devices, whereas probability and possibility devices occurred more often in the essays by higher proficient students. Moreover, the authors reported more instances in which Chinese learners of English chose to link themselves explicitly with their ideas rather than avoid personal involvement. Hyland (2012) further argued that Chinese learners of English are inclined to be straightforward and explicit in English and exhibit a strong desire to be connected to their texts. Hyland and Milton (1997) attributed these differences to either insufficient linguistic knowledge caused by different interpretations of equivalent semantic forms or to sociopragmatic violations brought about by an imperfect awareness of appropriate language use. Low (1996) referred to such inattention to or lack of awareness of important interpersonal features such as hedges and boosters as the lexical invisibility hypothesis and he explained that L2 learners might fail to process them adequately.

Gaskell et al. (1993) and Low (1996) explored the validity of the hypothesis with L1 participants, whereas Hyland (2000) examined it with L2 participants. Gaskell et al. (1993) looked into three areas in which they conducted research: how

the response alternatives affect responses, the impact of boosters in questions and memory effects. Their basic procedure was to ask a question with half the respondents being offered a response alternative set that discriminated at low frequencies and the other half being offered a response alternative set that discriminate at high frequencies. The results indicated that respondents in the two conditions resorted to different strategies to answer the questions. Furthermore, comparing response patterns for phrases with and without boosters (e.g., *very, extreme, extremely*) in a repeated measures design, they found that the boosters did not make any difference because respondents did not pay attention to the boosters and remember them. The only booster they remembered was ''extreme'' when added to ''physical pain'' because they are not decomposable into its parts and need to be processed holistically. They also explained that the repeated measures design forced the respondents to process ''extreme'' in the second question.

Low (1996) conducted a small-scale experiment with nine university students whose first language was English to investigate the validity of the findings by Gaskell et al. (1993). The participants answered a 30-item questionnaire consisting of 10 items about academic work, 10 items about life at university and 10 items about future plans. Three four-point bipolar scales were used in the questions, with different wording used in the three different scales: *strongly/mildly (dis)agree*, *completely/partially (dis)agree* and *strongly/mildly (dis)approve*. Low examined the participants' reactions to the adverbs *strongly*, *mildly*, *completely* and *partially* and analyzed whether the participants noticed six boosters (*very, extremely, far, full, never* and *consistently*) and two hedges (*seem* and *tend*) in the questionnaire items. The results revealed that four of the boosters received attention from half or more of the participants, unlike the findings by Gaskell et al. (1993), whereas the hedges and two boosters (*never* and *consistently*) were noticed much less, partially proving the validity of the claims by Gaskell et al. (1993). Although the findings by Low (1996) and Gaskell et al. (1993) were worthy of attention and carried important implications on hedges and boosters, their studies were carried out with L1 participants only, and the feasibility of applying the findings by Low (1996) and Gaskell et al. (1993) to L2 learning environments remained unexplored.

Hyland (2000) examined the applicability of the lexical invisibility hypothesis for hedges and boosters to the L2 learning environment in a study involving 14 Cantonese-speaking students studying in a BA program in English for Professional Communication. The data included taped interviews that sought to elicit the participants' awareness of hedges and boosters after reading a short passage and answering 15 true/false questions about the text. Participants also filled in a questionnaire with 15 multiple choice questions that focused more directly on their understanding of hedges and boosters. The 25 hedges and 10 boosters used as targets were drawn from the most common devices in a 500,000-word corpus of academic research articles from eight disciplines. The results provided strong evidence that the L2 learners attended to the boosters, whereas they consistently ignored hedges in the text and hedges seemed to be less visible.

In view of research by Hyland and Milton (1997) and Hyland (2000), it is clear that L2 learners' awareness of important markers of writer attitude, especially hedges, and an ability to recognize them in texts is crucial to the acquisition of rhetorical competence in any academic discipline. However, in their studies, the participants were students in Hong Kong where English is taught from kindergarten, with a number of other subjects also taught in English and schooling is a truly bilingual experience. On the other hand, in Japan, students do not start learning English until elementary school, 5th grade, at the earliest and schooling is monolingual in nature. Accordingly, the students in Hong Kong are more exposed to English and have more chances to use English. As a result, the students in Hong Kong consistently score among the highest on the TOEFL in Asia, whereas the students in Japan score among the lowest on the TOEFL in Asia (Educational Testing Service, 2014). Thus, the present study is interested in assessing the applicability of the tendencies found in the studies by Hyland and Milton (1997) and Hyland (2000) to Japanese learners of English and explores how Japanese learners of English deal with hedges and boosters in their academic writing.

5. Study 1

Study 1 follows the study of Hyland and Milton (1997) comparing the use of hedges and boosters by Japanese adult EFL learners and adult native speakers of English. Unlike the study by Hyland and Milton (1997) in which the Cantonese-speaking and English-speaking high-school aged students wrote English essays on different topics, in Study 1, all participants wrote English essays on the same topics so that there would be empirically valid equivalence between the writing by the Japanese college-aged adult EFL learners and the writing by adult native speakers of English. The following question is examined in Study 1:

> What are the relative differences in the use of hedges and boosters by Japanese EFL learners and native speakers of English in their English essays on the same topics?

5.1. Data and method

This first study draws data from the Nagoya Interlanguage Corpus of English (NICE) compiled by Sugiura (2014). The data consist of 552 essays on 11 topics written by 241 Japanese EFL learners (university students) (female = 170,

male = 71) and 38 native speakers of English (female = 11, male = 27). For the present study, 72 EFL learners (female = 58, male = 14) were chosen whose English proficiency was at the intermediate level, as defined by TOEIC scores of between 400 and 685. The 72 essays written by the EFL learners on 10 topics were compared with the 38 essays written by 38 native speakers of English on the same topics, namely *the death penalty, divorce, money, school education, sports, suicide, teenagers, recycling, violence on TV* and *water pollution.* There were only 38 native speakers of English (37 college graduates and one high school graduate) participating in the NICE. The ages of the native speakers were varied ranging from 20 to 70 years old. All participants were instructed to write freely without regard to any particular genre on the same topics within an hour and they were not allowed to use any dictionaries.

5.2. Analysis

The essays were converted into an electronic corpus of 48,175 words (22,386 words from the EFL learners and 25,789 words from the native speakers of English). Lexical expressions of hedging and boosting were analyzed using AntConc 3.4.3 (Anthony, 2014) by the researcher and statistical analyses were conducted using SPSS Version 19.0 (2010) by the same researcher. First, based on the definitions of a hedge and a booster in Section 1, adjectives, adverbs, auxiliaries, nouns, and verbs used by the EFL learners and the native speakers were divided into hedges and boosters and then frequencies of them were calculated. Second, based on the epistemic categories in Section 2, the frequencies of the certainty, probability, and possibility items were calculated. Third, the participants' indication of subjective and objective orientation along the subjective/objective dimension in Section 3 was investigated.

Of note is the considerable difficulty of mapping a precise degree of conviction onto particular terms in every instance. Modal expressions are rarely fixed with respect to particular meanings because their semantic values are always subject to pragmatic constraints that can alter their meanings. As an example of this ambiguity, *believe* and *think* can be either hedges or boosters depending on the context. Thus, these two verbs were excluded from the list used in the corpus analysis for this study, but the use of *believe* and *think* will be examined from the perspective of a subjective explicit orientation. Moreover, although *assume*, *hypothesize* and *speculate* are considered to belong to the probability item group, the present study included them in the possibility item group because their degree of sureness resembles that of the possibility items, as proposed by Biber et al. (1999), Hyland and Milton (1997), Halliday and Matthiessen (2014), Quirk et al. (1985) and Huddleston and Pullum (2002).

5.3. Results of Study 1

5.3.1. Frequencies of hedges and boosters

Table 2 shows the most frequently occurring hedges and boosters in the corpus, with auxiliaries such as *would*, *could*, *may* and *might* accounting for 32% of the hedging devices in the essays by the EFL learners and with *would*, *may*, *could, should* and *can* accounting for 34% of the hedging devices used by the native speakers. Regarding the boosting devices, *will* and *know* make up 38% of the boosting devices used by the EFL learners and 27% of the boosting devices used by the native speakers. Both groups made substantial use of auxiliaries, particularly *will* and *would* to convey epistemic meanings. In addition, *may* appeared in the EFL learners' and the native speakers' essays as a hedging device but occurred about twice as often in the EFL learners' essays.

Table 2
Most frequent hedges and boosters in the corpus.

EFL learners				Native speakers			
Hedges	Frequency	Boosters	Frequency	Hedges	Frequency	Boosters	Frequency
Would	32 (1.43)	Will	89 (3.98)	Would	61 (2.37)	Will	109 (4.23)
Could	31 (1.38)	Know	54 (2.41)	May	60 (2.33)	Know	35 (1.36)
May	30 (1.34)	Especially	27 (1.21)	Seem	46 (1.78)	Fact	27 (1.05)
Consider	21 (0.94)	Always	18 (0.80)	Could	36 (1.40)	Always	24 (0.93)
Often	18 (0.80)	Never	18 (0.80)	Most	33 (1.28)	Never	22 (0.85)
Almost	14 (0.63)	Really	15 (0.67)	Should	31 (1.20)	Certain	18 (0.70)
Might	12 (0.54)	True	14 (0.63)	Can	26 (1.01)	Result	16 (0.62)
Usually	10 (0.45)	Find	11 (0.49)	Consider	17 (0.66)	Find	14 (0.54)
Hope	9 (0.40)	Show	11 (0.49)	Feel	17 (0.66)	Really	14 (0.54)
Maybe	9 (0.40)	Would	8 (0.36)	Sometimes	16 (0.62)	Especially	13 (0.50)

Note. The figures in parentheses show the number per 1000 words.

Table 3
Grammatical distribution of hedges and boosters.

Category	EFL learners		Native speakers	
	Hedges	Boosters	Hedges	Boosters
Adjectives	17 (0.76)	42 (1.88)	103 (3.99)	104 (4.03)
Adverbs	90 (4.02)	100 (4.47)	135 (5.23)	136 (5.27)
Auxiliaries	113 (5.05)	116 (5.18)	223 (8.65)	132 (5.12)
Nouns	23 (1.03)	21 (0.94)	34 (1.32)	64 (2.48)
Verbs	83 (3.71)	102 (4.56)	116 (4.50)	103 (3.99)
Totals	326 (14.56)	381 (17.02)	611 (23.69)	539 (20.90)

Note. The figures in parentheses show the number per 1000 words.

The results shown in Table 3 indicated that the EFL learners used more boosters than hedges, whereas the native speakers used more hedges than boosters. Worth particular attention is the fact that hedges by the native speakers exceeded hedges by the EFL learners by nearly 1.9–1, t (8) = 2.94, p = .04 < .05, while there were no significant differences between boosters by the EFL learners and boosters by the native speakers, t (8) = 2.14, p = .10 > .05, and there are marked frequency differences in the use of adjectives, adverbs and auxiliaries between the EFL learners and the native speakers. The quantitative results support the view that non-native speakers tend to express themselves more assertively, whereas native speakers are inclined to express themselves more tentatively. Hedging and boosting by both groups was accomplished most frequently through the use of auxiliaries and least frequently through the use of nouns.

The present study has thus far examined the overall frequency of hedges and boosters used by the EFL learners and the native speakers, but to gain a fuller picture of the tendencies in the use of hedges and booster, a precise breakdown of hedges and boosters used by the EFL learners and the native speakers needs to be examined.

Table 4 shows that although the frequencies of the certainty items are similar between the EFL learners and the native speakers, discrepancies exist in frequencies of the probability and possibility items. That is, in spite of the fact that both EFL learners and the native speakers prefer to use auxiliaries for probability and possibility, the EFL learners did not use as many of these auxiliaries as the native speakers did.

My answer would be yes to the former and no to the latter. (EFL)
She would feel sad. (NS)

In addition, the EFL learners did not use epistemic adjectives and adverbs for probability and possibility as frequently as the native speakers did.

So many young, perhaps including me, don't know nationalism. (EFL)
With high enough grades in the proper courses, entrance in to any university in Canada is possible. (NS)

Table 4
Frequency of items in main epistemic categories.

EFL learners						Native speakers					
Certainty	Freq.	Probability	Freq.	Possibility	Freq.	Certainty	Freq.	Probability	Freq.	Possibility	Freq.
Clear	2 (0.09)	Probably	4 (0.18)	Possible	6 (0.27)	Clear	8 (0.31)	Likely	17 (0.66)	Possible	17 (0.66)
Obvious	1 (0.04)	Seem	6 (0.27)	Potential	1 (0.04)	Distinct	1 (0.04)	Probably	8 (0.31)	Potential	1 (0.04)
Certainly	3 (0.13)	Suggest	7 (0.31)	Perhaps	1 (0.04)	Evident	1 (0.04)	Seem	46 (1.78)	Perhaps	10 (0.39)
Obviously	1 (0.04)	Should	8 (0.36)	Could	31 (1.38)	Obvious	6 (0.23)	Suggest	1 (0.04)	Possibly	4 (0.16)
Claim	2 (0.09)	Would	32 (1.43)	May	30 (1.34)	Certainly	5 (0.19)	Should	31 (1.20)	Potentially	3 (0.12)
Know	54 (2.41)			Might	12 (0.54)	Clearly	2 (0.08)	Would	61 (2.37)	Assume	1 (0.04)
Find that	11 (0.49)					Obviously	1 (0.04)			Could	36 (1.40)
Will	99 (4.42)					Undoubtedly	3 (0.12)			May	60 (2.33)
						Claim	1 (0.04)			Might	9 (0.35)
						Know	35 (1.36)				
						Find that	14 (0.54)				
						Will	111 (4.30)				
Totals	173 (7.73)		57 (2.55)		80 (3.57)		189 (7.33)		164 (6.36)		141 (5.47)

Note. Freq. = frequency; the figures in parentheses show the number per 1000 words.

5.3.2. Subjectivity and objectivity

The EFL learners were more likely to emphasize subjectivity, which indicates that the EFL learners were more likely to assume personal responsibility for their epistemic judgment by more frequent use of cognitive verbs such as *think* and were more willing to involve themselves in the judgment by using personal attribution.

This result coincides with the findings of Hyland and Milton (1997), showing more instances in which the EFL learners chose to link themselves explicitly with their ideas rather than avoiding personal involvement. As the EFL learners were asked to write academic essays rather than academic articles, the conventions of academic discourse did not prevent the EFL learners form expressing their personal views:

I think maybe I have a special power. (EFL)
I think adults might forget their own school education. (EFL)
I think that applying the death penalty gives relatives and friends of murder victims a vitally important sense of closure. (NS)
I believe that the death penalty should not be used because it is important to kill a person who may potentially be innocent of a crime. (NS)

Personal pronoun *I* did not occur with the epistemic verb *believe* to indicate the EFL learners' strong belief, but it occurred frequently with the epistemic verb *think* in situations where the EFL learners tried to strengthen the force of an argument or to weaken a claim by hedging its generalizability. Their attempts of using *I think* often result in attenuating their claims because they are accustomed to using this expression as a weak form of stance marker in their spoken discourse. The native speakers of English, on the other hand, did not use *I think*/*I believe* very frequently, but when they used them, they clearly indicated their personal beliefs and thoughts. The overuse of *I think* by Japanese learners of English has been reported by different researchers (e.g., McClostie, 2008; Pech-Tyson, 1998) and their finding showed that native speakers of English seldom used *I think*, while the EFL learners used *I think* very frequently (Tables 1 and 5).

Table 5
Disciplinary differences in the use of the cognitive verb, *think, believe*.

	EFL learners	Native speakers
I think	164 (7.33)	15 (0.58)
I believe	0	17 (0.66)

Note. The figures in parentheses show the number per 1000 words.

By contrast, the objectivity intention can be more easily realized by resorting to impersonal strategies such as faceless subjects, abstract subjects, agent-less passives and dummy *it*:

It is possible that our nationalism is not *Hinomaru* of *Kimigayo*. (EFL)
It is clear that this situation must be changed to make better. (EFL)
It is unlikely that the student will gain native fluency. (NS)
It is clear that education has evolved over many years to take on its many different modern forms. (NS)
Statistics show that a holder of a college degree both earns more and is more likely to be employed than a person who merely finishes high school. (NS)

As Table 6 shows, the EFL learners and the native speakers appeared not to resort to the objective and impersonal expressions that are often seen in natural science articles.

Table 6
Differences in personal and impersonal strategies.

		EFL learners	Native speakers
Subjective explicit	Totals	209 (9.34)	63 (2.44)
Objective explicit (with dummy *it*)	Totals	15 (0.67)	27 (1.05)
Faceless subjects, abstract subject, agentless passives (excluding dummy *it*)	Totals	0 (0.00)	4 (0.16)

Note. The figures in parentheses show the number per 1000 words.

Objectivity is a value in natural science that informs how science is practiced and how scientific truths are discovered. For this reason, in natural science articles writers attempt to eliminate personal biases or commitments toward propositional content. However, the present study did not address academic articles but rather academic essays, which allow the writers to explore and reflect on their personal experiences or ideas, providing readers with evidence for a particular point or claim. Therefore, the differences in genre between the natural science articles and the academic essay appear to have influenced the few cases in which objective and impersonal strategies were used.

6. Study 2

Study 2 follows the study of Hyland (2000) examining the applicability of the lexical invisibility hypothesis for hedges and boosters and exploring whether Japanese learners of English are able to recognize the important pragmatic roles played by hedging and boosting devices and the subjective/objective dimension of the grammatical metaphor in academic essays, building on the results of Study 1. The following question is looked into in Study 2:

> Are Japanese learners of English able to recognize the certainty, probability and possibility items of lexical metaphor and subjective/objective dimension of the grammatical metaphor in academic essays?

6.1. Participants

Twenty-seven students (female = 2, male = 25) at a Japanese university participated in the second study. The participants were non-English majors, studying in the College of Science and Engineering and were not aware that English hedges and boosters were the target features of the study. The participants' English proficiency levels were considered to be at the intermediate level, as defined by a TOEIC score of 400–700. The participants' first language was Japanese, and their average age was 20 years old. All participants had studied English for eight years at schools in Japan. In addition, six native speakers of English (female = 2, male = 4) also participated in the study and their data was used as the baseline data. The native speakers' data was relatively uniform and consistent.

6.2. Target features

The results of Study 1 indicated that although similarity is apparent in the frequencies of the certainty items used by the EFL learners and the native speakers, discrepancies in the frequencies of the probability and possibility items. The EFL learners did not use the items expressing probability and possibility as frequently as the native speakers did. In other words, the EFL learners did not moderate their claims as fully as the native speaks did. Thus, Study 2 employed the items

Table 7
Certainty/probability/possibility items.

Certainty	Probability	Possibility
Clearly	Likely	Possible(ly)
Obviously	Probable(ly)	Could
Undoubtedly	Appear	May
Claim	Seem	Might
Confirm	Suggest	Assume
Demonstrate		Hypothesize
Establish		Speculate
Find		
Know		
Show		
Confirmation		
The fact that		

Table 8
Subjective/objective dimension of the grammatical metaphor.

		Subjective	Objective
Explicit	Hedge	I suppose that...	It is likely that...
	Booster	I know that...	It is certain that...

expressing probability and possibility as the target items to explore whether the Japanese learners of English recognized the important roles played by hedging and boosting devices in essays.

Furthermore, Study 2 examined whether the Japanese learners of English are aware of the subjectivity and objectivity of hedges and boosters.

6.3. Testing instrument

The data were collected using an acceptability judgment test and a questionnaire that sought to assess the participants' awareness of hedges and boosters. Before the participants completed the acceptability judgment test and the questionnaire, they were required to read a short passage about green tea (see Appendix A). The passage was chosen for the relevance of the topic to the participants who would be familiar with most of the vocabulary. The passage was modified slightly and shortened to a manageable length (370 words). The passage, the acceptability judgment test and the questionnaire contained 14 hedges and 12 boosters in total. The hedges and boosters were drawn from the common hedges and boosters found in a 369,905 word corpus of academic research articles from eight disciplines. A native speaker of English proofread the passage and ensured that the passage itself was not unusual in any way. The acceptability judgment test (see Appendix A) required the participants to read 15 statements related to the passage and then revise them or leave them as they were in accordance with the contents of the passage. The participants were asked to circle the words they used to make their revisions. These sentences were intended to focus the participants' attentions on hedges, boosters and the subjective–objective dimension of hedging. The participants were given 30 min to complete the acceptability judgment test. As the test consisted of 15 statements, the maximum score was 15 points. As in Hyland's (2000) study, the participants were then asked to answer the questionnaire (see Appendix A). The data from the questionnaire were intended to supplement the results of the acceptability judgment test in exploring the participants' awareness of the target hedges and boosters. The questionnaire used 12 statements from the passage and simply asked the participants to rate each statement according to whether they thought the writer was ''completely certain,'' ''fairly certain,'' or ''uncertain'' about the claim or whether they did not know. They were also asked to circle the words they used to make their choice. The participants were given 15 min to complete the questionnaire. As there were 12 statements in the questionnaire, the maximum score was 12 points.

6.4. Analysis

With respect to internal consistency, the average Cronbach alpha reliability estimates for the acceptability judgment test and the questionnaire were calculated to be .751 and .219, respectively, indicating reasonable internal consistency for the acceptability judgment test but low internal consistency for the questionnaire due to low variability in the participants' performance the present study tested and also the fact that the poor performance on particular questions about probability and possibility items was common to almost all participants.

Because of the small number of cases, content validity was assessed rather than criterion and construct validity. To ensure content validity, situations of the two testing instruments were carefully planned and matched to a theoretical framework based on certainty, probability and possibility variables.

Statistical analyses using the SPSS program showed no violation of the normality assumption. Parametric tests were used to analyze the data from the acceptability judgment test and the questionnaire. The overall alpha level was set at .05. With respect to maintaining an approximate experiment-wide .05 alpha level, a Bonferroni adjustment needed to be made before the post hoc tests. Because three comparisons were made (for the certainty, probability and possibility items), the overall significance level of .05 had to be divided by three for individual statistical decisions before the post hoc *t*-tests. The scores for the acceptability judgment test and the questionnaire were graded by the researcher and were analyzed using the SPSS 19.0 (2010) by the same researcher. The participant ratings were compared with English native speakers' baseline data. The following section summarizes the results for the acceptability judgment test and the questionnaire.

6.5. Results of Study 2

6.5.1. Results from the acceptability judgment test

Results of a one-way ANOVA showed a significant main effect for three items (the certainty, probability and possibility items), F (2, 78) = 3.11, p = 3.83E−35 < .05. The post hoc *t*-tests for the main effect of treatment indicated the following contrasts: (a) the scores on the certainty items were significantly higher than the scores on the probability items [t (26) = 19.22, p = .000 < .017] and the possibility items [t (26) = 34.39, p = .000 < .017]; (b) significant differences existed between the scores on the probability and the possibility items [t (26) = −3.12, p = .004 < .017] (Tables 7 and 8).

As Table 9 shows, like the native speakers of English, most of the participants were able to correctly notice all certainty statements, but all of the participants had difficulty in recognizing the probability and possibility statements. The

Table 9
Descriptive statistics for the acceptability judgment test.

Item	Score	Mean	SD
Certainty	5	4.33 (5.00)	0.23 (0.00)
Probability	5	0.74 (4.83)	0.89 (0.41)
Possibility	5	0.26 (4.00)	0.43 (1.26)

Note. The figures in parentheses show the baseline data provided by the native speakers of English.

participants demonstrated the greatest difficulty in recognizing the possibility statements, and they were not able to distinguish the probability statements from the possibility statements. For example, although most of the participants were able to notice auxiliaries and verbs such as *may, might, seem* and *suggest* in Statements 4, 8, 10, 11 and 13, what counts as *possible(-ly)* or *probable(-ly)* was not clear to them, and only two participants were able to correctly replace *possibly* with *probably.* However, the participants confounded the meanings of *possibly* and *probably* in Statement 14. Also, few noticed *could* in Statement 9 and only two participants were able to correctly replace *can* with *could*. Furthermore, participants seemed to have the utmost difficulty with Statements 6, 7, 13 and 14 that concerned the subjective–objective dimension of the grammatical metaphor. Only two participants were able to pinpoint differences in subjective–objective and explicit–implicit orientations.

6.5.2. Results from the questionnaire

Results of a one-way ANOVA showed a significant main effect for three items (the certainty, probability and possibility items), F (2, 78) = 81.49, p = 7.85E−20 < .05. The post hoc t-tests for the main effect of treatment indicated the following contrasts: (a) the scores on the certainty items were significantly higher than the scores on the probability items [t (26) = 9.50, p = 6.1E−10 < .017] and the possibility items [t (26) = 15.62, p = 9.98E−15 < .017]; (b) significant differences existed between the scores on the probability and the possibility items [t (26) = −2.80, p = .0010 < .017].

As Table 10 indicates, like the native speakers of English, almost all of the participants marked the certainty statements correctly like the native speakers of English, but they had difficulty in assigning an appropriate degree of certainty to the probability and possibility statements. As in the acceptability judgment test, they exhibited the greatest difficulty in assigning an appropriate degree of sureness to the possibility statements. These results are consistent with the findings of Hyland (2000), and although the participants were able to notice auxiliaries and verbs such as *may, might, assume, seem, appear, hypothesize* and *suggest* in Statements 1, 5, 6, 7, 9, 10 and 11, they had difficulty noticing and grasping the meanings of *could*, *possible(-ly)* and *probable(-ly)* in particular, in Statements 7, 10, 11 and 12. Thus, the participants who circled either *possible* or *probable* confounded the meanings of the two. Only two participants were able to detect the difference in their meanings and were able to correctly comprehend the meaning of *could* as a possibility item. The data from the questionnaire confirmed the finding from the acceptability judgment test that the participants did not recognize the different pragmatic roles of the possibility and probability items, especially *could*, *possible(-ly)* and *probable(-ly)*, which add different degrees of tentativeness and caution to academic statements.

Table 10
Descriptive statistics for the questionnaire.

Item	Score	Mean	SD
Certainty	4	3.85 (4.00)	0.13 (0.00)
Probability	4	1.74 (3.33)	1.28 (1.03)
Possibility	4	1.00 (3.50)	0.77 (0.84)

Note. The figures in parentheses show the baseline data provided by the native speakers of English.

7. Discussion

The results of Study 1 showed that although the frequencies of the certainty items are similar between the EFL learners and the native speakers, the EFL learners employed more certainty items than probable or possibility items, whereas native speakers of English used more probable and possible items than certainty items. Moreover, the results of Study 1 indicated more instances in which the EFL learners chose to link themselves explicitly with their ideas rather than avoid personal involvement. The fact that the EFL learners did not moderate their claims as thoroughly as the native speakers

did could have been due to their inadequate linguistic knowledge, or they might have been unaware of the important pragmatic roles played by the hedging devices and objective dimension of the grammatical metaphor. Accordingly, although the EFL learners may express themselves more indirectly in Japanese, inadequacy with respect to their linguistic knowledge or a lack of awareness of the pragmatic roles of the hedging devices may have influenced their use of an inappropriate degree of directness. In order to find a probable cause of the results of Study 1, Study 2 explored whether the EFL learners are able to recognize and categorize the pragmatic roles played by hedging devices and the subjective–objective dimension of the grammatical metaphor.

The results of Study 2 indicated that although the certainty items were more visible to the EFL learners, the learners had greater difficulty in identifying and categorizing the pragmatic roles of the possibility and probability items and misapprehended the meanings of the two. The results also suggested that participants had difficulty with the subjective–objective dimension of the grammatical metaphor. Only a few participants were able to pinpoint differences in subjective–objective orientations possibly due to their inadequate linguistic knowledge of the important pragmatic roles played by the subjective–objective dimension of the grammatical metaphor.

The idea as to the EFL learners' knowledge about the probability and possibility items through spatial relations is necessarily speculative as no information relating to the psycholinguistic processing involved in the test is available. However, in view of the results of Study 1 and Study 2 that the EFL learners did not attenuate their claims as thoroughly as the native speakers did and the EFL learners failed to recognized and distinguish possibility items from probability items according to the degree of or the distance from the writer's confidence, from the standpoint of the proximal–distal metaphor, it could be assumed that the learners' concepts about the degree of sureness connected with the possibility and probability items, especially *could* and *possible(-ly)* and *probable(-ly)*, might not be deeply entrenched in the participants' knowledge about their spatial relations (Ohori, 2002; Ohori and Bo Sen, 2012). In other words, it is likely that the EFL learners perceive that certainty items are used as being closer to the writer's confidence, indicating a higher degree of sureness, but it seems that the EFL learners do not perceive that probability and possibility items are used differently as being further from the writer's confidence at different distances, indicating a lower degree of sureness.

8. Conclusion

The corpus-based analysis of Study 1 suggested that the EFL learners tended toward confident prediction, whereas native speakers preferred to express themselves more tentatively. The EFL learners relied on a limited variety of hedging devices to express their epistemic commitment toward propositions. The results of Study 2 indicated that although the participants were able to notice certainty statements correctly, they had difficulty in recognizing the probability and possibility statements, having the greatest difficulty with the possibility statements. The EFL learners also had trouble recognizing the differences in subjective–objective expressions of hedging and boosting.

The pedagogical implications of these findings are that teachers should be aware that EFL learners' knowledge about the different degrees of sureness attached to expressions of possibility and probability of lexical metaphor and the different degrees of assertiveness attached to subjective–objective dimension of the grammatical metaphor may not be firmly established and easily accessible, which results in the use of fewer hedging and tentative expressions. Thus, teachers should attempt to make the certainty, probability and possibility items easier to learn by showing that certainty items are placed as being closer to the writer, denoting a higher degree of the writer's confidence in propositions, whereas probability and possibility items are placed differently, as being further from the writer at different distances, meaning two different levels of lower degrees of the writer's confidence in propositions. In addition, teachers should help EFL learners become more aware of the importance of claims being presented provisionally or objectively rather than assertively or subjectively in order to gain ratification from powerful peer readers in academic writing. As spatial relations in cognition plays a significant role in human life (Ohori, 2002; Ohori and Bo Sen, 2012), teaching the markers of the certainty, probability, and possibility items of the lexical metaphor through spatial relations will prove of great value in entrenching EFL learners' knowledge about them and improving their learners' use of assertive and tentative expressions. With greater numbers of L2 students learning how to write effective academic articles in English, teaching them to have an even balance of the writer's assertiveness and tentativeness markers must become a higher priority and receive more consideration.

Regarding the limitations of the present studies, the findings here are related to academic essays written by the EFL learners and the native speakers of English and they wrote their essays freely on the same topics without regard to any particular genre, but the applicability of the findings to academic articles with article genres written by EFL learners is not clear. Accordingly, future research should address how hedges and boosters are dealt with in academic articles written by EFL learners. Furthermore, the results of the questionnaire in Study 2 need to be treated with some caution due to the low reliability estimate associated with the EFL learners' inadequacy in their linguistic knowledge and their random answering on that account. As a questionnaire with a reasonable degree of reliability would have been ideally suited to confirm the

findings of the acceptability judgment test, a comprehensive set of screening methods for careless responding should be provided for future research.

Despite the shortcoming, the two studies presented here contribute to our understanding of the marked characteristics of the use of hedges and boosters in academic essays by EFL learners and native speakers of English and to our understanding of EFL learners' awareness of the pragmatic roles of hedges and boosters in academic essays. In addition, the unique challenges and opportunities faced in conducting comparative studies of hedges and boosters in academic essays written by EFL learners and native speakers of English serves to expand scholarship not only in the area of intercultural phenomena but also in the wider field of applied linguistics.

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Appendix A

Please read the passage and then follow the directions that follow.

Green tea is getting much attention these days because many researchers assume that drinking a few cups of green tea a day may help prevent cancers and many studies demonstrate that drinking green tea is good for the health. These provide scientific confirmation for the legendary saying: ''green tea is a miraculous medicine for the maintenance of health.'' Many studies also establish the fact that green tea is loaded with polyphenols that protect cells from the bodily changes caused by a chemical reaction that can lead to cancer and heart disease.

In recent studies green tea and its component, catechin, have seemed to reduce the growth of cancer and have appeared to prevent the actual production of cancer. Even though these results have been gained from animal studies or pure laboratory tests, researchers speculate it highly significant that tea catechin has the ability to prevent cancer.

Other studies suggest that the catechin green tea may act to limit excessive blood cholesterol, has the ability to prevent increased blood pressure, and can lower blood sugar. Recent Japanese studies have shown that green tea likely contains a powerful antioxidant which researchers hypothesize could help control aging. Other research has clearly found that the antibacterial function of green tea catechin is effective in preventing food poisoning. Moreover, dental research suggests that green tea catechin both slows down the formation of plaque by oral bacteria and kills the bacteria themselves. Obviously, green tea can also strengthen teeth and will help prevent cavities because of the fact that it contains natural fluorine.

Green tea, with its sweet aroma and fresh taste, has been loved since its introduction to Asian countries centuries ago. But modern research has finally started to remove the veil concealing its true value as a functional food. It is possible to say that green tea just might be a truly miraculous medicine with an extraordinary power to prolong life. If you already consume caffeine in one form or another, it is probably a good idea that you drink green tea all the time, Instead of having a soda in the daytime or a cup of coffee at night, doctors will clearly advise that you treat yourself to a cup of green tea.

A. In view of the contents of the passage, if necessary, please revise the sentence below underlining the parts you revise and if not necessary, please leave it as it is.

(1) Many researchers assume that drinking a few cups of green tea a day helps prevent cancers.
(2) Many studies clearly confirm the legendary saying: ''green tea is a miraculous medicine for the Maintenance of health.''
(3) Many studies prove the fact that green tea is loaded with polyphenols.
(4) According to the animal studies, catechin reduces the growth of cancer.
(5) In the pure laboratory tests, catechin may have prevented the actual production of cancer.
(6) It is probable that tea catechin has the ability to prevent cancer.
(7) It is possible that catechin limits excessive blood cholesterol.
(8) According to recent Japanese studies, green tea might contain a powerful antioxidant.
(9) A powerful antioxidant can help control aging.
(10) Other research has suggested that the antibacterial function of green tea catechin may be effective in preventing food poisoning.
(11) Dental research claims that green tea catechin slows down the formation of plaque by oral bacteria.
(12) We know for certain that green tea contains natural fluorine which can strengthen teeth and help prevent cavities.
(13) Modern research reflects the fact that green tea is a truly miraculous medicine with an extraordinary power to prolong life.

(14) If you already consume caffeine in one form or another, it is possibly a good idea to try to change to green tea at least some of the time.
(15) The doctors undoubtedly advise that you treat yourself to a cup of green tea.

B. Please mark each statement according to whether you think the writer wants to show the writer is ''completely certain'' (C), ''fairly certain'' (F), or ''uncertain'' (U), about the results or claim in the sentence. When you mark C, F or U, please underline the words you used to make your choice. You can just respond with ''I don't know'' (D) if you cannot decide.

(1) Many researchers assume that drinking a few cups of green tea a day may help prevent cancers._____
(2) Many studies demonstrate that drinking green tea is good for the health._____
(3) Many studies clearly confirm the legendary saying: ''green tea is a miraculous medicine for the maintenance of health.'' _____
(4) Many studies establish the fact that green tea is loaded with polyphenols. _____
(5) Green tea and catechin have seemed to reduce the growth of cancer. _____
(6) Green tea and catechin have appeared to prevent the actual production of cancer. _____
(7) Researchers hypothesize a powerful antioxidant could help control aging. _____
(8) Other research has clearly found that the antibacterial function of green tea catechin is effective in preventing food poisoning. _____
(9) Dental research suggests that green tea catechin slows down the formation of plaque by oral bacteria. _____
(10) It is possible to say that green tea just might be a truly miraculous medicine. _____
(11) It is possible to say that green tea might have an extraordinary power to prolong life. _____
(12) If you already consume caffeine in one form or another, it is probably a good idea that you drink green tea all the time. _____

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